

Retrieval-Augmented Generation Approach for SPM Extraction

*Extraction of SPM-Protein Interactions from Biomedical
Literature*

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Challenges in Biomedical Literature



Exponential Growth

PubMed indexes over 30 million papers, with thousands added daily. Manual curation becomes impossible.



Keyword Limitations

Simple keyword searches miss context and fail to capture semantic meaning in complex biological concepts.



Synonym Complexity

Same molecules appear under multiple names and abbreviations across different research communities.



Relationship Extraction

Meaningful biological relationships are buried in unstructured text, requiring semantic understanding.

Need: Semantic understanding combined with structured, automated extraction of meaningful relationships from vast literature repositories.

Why Specialized Pro-Resolving Mediators?

What are SPMs?

Specialized Pro-Resolving Mediators are bioactive lipid molecules that actively regulate inflammation and immune response. Unlike traditional anti-inflammatories that suppress immune activity, SPMs promote resolution and tissue repair.

Clinical Significance

- Chronic inflammatory diseases
- Tissue repair mechanisms
- Drug development targets
- Immune modulation therapies

The Challenge

SPM-related information is scattered across thousands of papers using inconsistent terminology and nomenclature.



Project Objectives

- Expand SPM terminology using PubChem synonyms
- Retrieve relevant biomedical literature from PubMed
- Enable semantic search using embedding techniques
- Develop a RAG-based chatbot for question answering
- Extract SPM–protein interactions with supporting evidence
- Visualize structured knowledge using a graph database

Retrieval-Augmented Generation Framework



Information Retrieval

Query relevant documents from large corpus using semantic embeddings and vector similarity

Language Model Processing

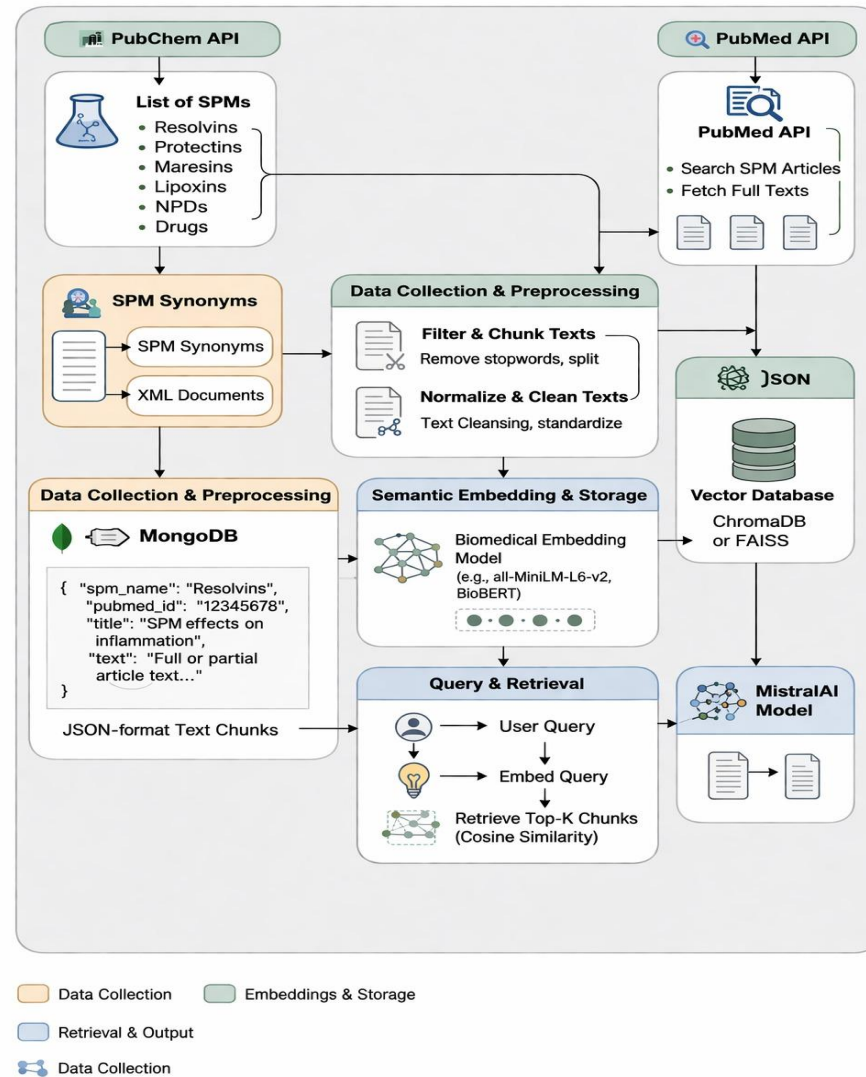
Large language model analyzes retrieved evidence to generate contextual answers

Evidence-Based Reasoning

Answers grounded in actual scientific papers, reducing hallucination and improving accuracy

Advantages: More accurate answers, source-based reasoning, reduced hallucination, transparent citations, and improved trustworthiness.

System Architecture Overview



Methodology Pipeline

01

SPM Synonym Extraction

Query PubChem for all known SPM names, abbreviations, and chemical identifiers

03

Data Storage

Store full-text papers, metadata, and structured information in MongoDB

05

Vector Indexing

Build FAISS index for efficient approximate nearest neighbor search

07

Interaction Extraction

LLM processes retrieved evidence to extract SPM–protein relationships

02

Paper Retrieval

Search PubMed and PMC using expanded synonym list with relevance filters

04

Embedding Generation

Generate sentence-transformer (SBERT) embeddings for all text chunks

06

RAG Retrieval

Retrieve top-k relevant document chunks for each query using semantic similarity

08

Knowledge Graph

Store structured interactions in Neo4j for visualization and querying

Semantic Search Mechanism

Vector Embeddings

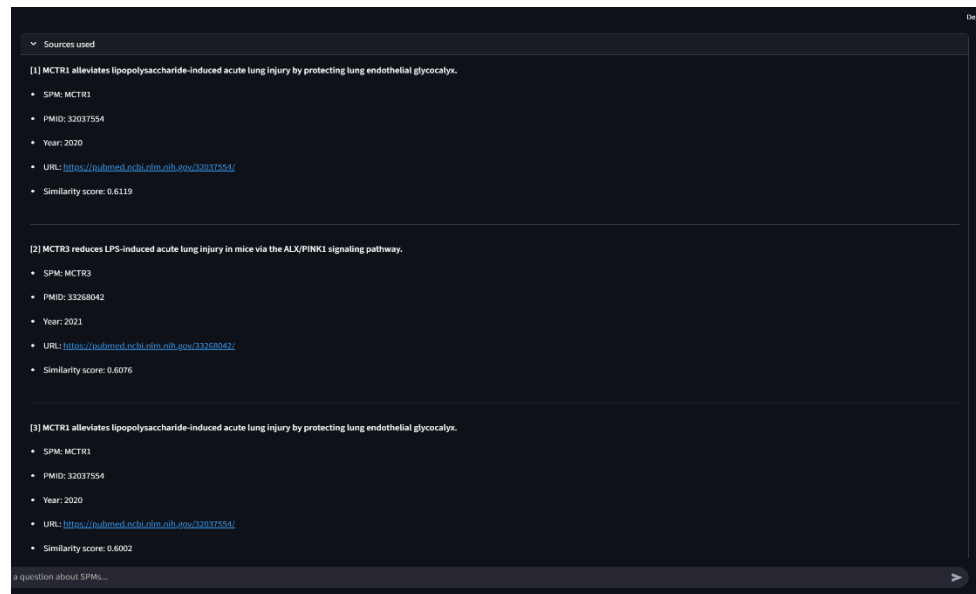
- Convert chunks into high-dimensional vectors using SBERT.
- Similar meanings produce similar vectors regardless of exact wording.

Cosine Similarity

Calculate vector similarity to find most relevant document chunks for any given query.

Advantages Over Keywords

- Captures semantic meaning
- Handles synonyms naturally
- Context-aware matching
- Improved relevance ranking



RAG Chatbot & Interaction Extraction

Query Processing

User question converted to embedding vector using same SBERT model

Document Retrieval

FAISS finds top-k most relevant text chunks from indexed corpus

LLM Generation

Mistral processes retrieved evidence to generate citation-backed answers

Extracted Information

SPM Identifier

Standardized molecule name with PubChem ID

Target Protein

UniProt identifier and protein name

Interaction Type

Relationship classification: binds, activates, inhibits, etc.

Evidence Sentence

Exact text supporting the interaction claim

Source Citation

PMID and document location for verification

Knowledge Graph Visualization



SPM Nodes

Specialized Pro-Resolving Mediators represented as nodes with chemical properties and classification



Protein Nodes

Target proteins shown with UniProt identifiers, functions, and biological pathways



Publication Nodes

Source papers linking interactions to original evidence with citation metadata



Relationship Edges

Interaction types connecting SPMs to proteins with supporting evidence references

```
neo4j$ MATCH (s:SPM)-[r:INTERACTS_WITH]->(p:Protein) WITH s, count(r) AS
```

Graph Table RAW



Results overview

Nodes (44)



* (44)

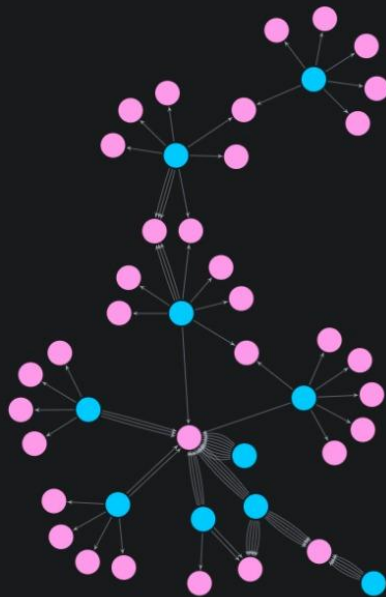
Protein (34)

SPM (10)

Relationships (81)

* (81)

INTERACTS_WITH (81)



Started streaming 81 records after 184 ms and completed after 235 ms.

Key Results

57

SPMs Processed
Specialized Pro-Resolving Mediators

9.7K

Papers Collected
From PubMed and PMC

54K

Indexed Chunks
Text segments for retrieval

140

Interactions Extracted
SPM–protein relationships

61

Unique Proteins
Identified targets

~11

Synonyms
Identified average Synonyms per SPM

Challenges & Future Directions

Current Challenges

- Synonym ambiguity in literature
- Limited full-text access (only 20% available)
- Data inconsistencies across sources
- Chunk size trade-offs
- Retrieval ranking optimization

Future Improvements

- Biomedical-specific LLMs (BioBERT, SciBERT)
- Improved evaluation metrics
- Relation normalization
- Hybrid retrieval (semantic + keyword)
- Advanced graph analytics

Conclusion

The proposed RAG pipeline effectively bridges the gap between unstructured biomedical data and structured knowledge enabling discovery of meaningful SPM–protein interactions.

Thank You! 😊